MSc Data Analytics (Jan 23/Sept 23/Jan 24 intake)- Research Project

MSc Cloud Computing – Research Project

MSc Fintech, MSc Cybersecurity, MSc AI, MSc AI for Business – Practicum Part 2/ Internship Part 2

MSc Data Analytics (Sept 24 intake and onwards) – Research Practicum Part 2/ Internship Part 2

Module Overview

Module Overview

- **Module name:** **Research Project**
- **Credit Rating:** 25
- **Credit Level:** 9
- **Contact Hours:** Timetabled weekly sessions

  E.g. 2-4 hours based on group size
- **Module Lecturer:** **Check Timetable for your group**

MSc Data Analytics
(Jan 23/Sept 23/Jan 24 intakes only)

MSc Cloud Computing

Research in Computing → Research Project

Module Overview

- **Module name:** **Practicum Part 1 & 2 / Internship Part 1 & 2**
  Credit Rating: 5+ 25
- **Credit Level:** 9
- **Contact Hours:** Timetabled weekly sessions

  Practicum/ Internship Part 2 - E.g. 2-4 hours based on group size
- **Module Lecturer:** **Check Timetable for your group supervisor**

MSc AI
MSc AI for Business
MSc Cybersecurity
MSc Fintech

Practicum Part 1/ Internship Part 1 → Practicum Part 2/ Internship Part 2

Module Overview

- **Module name:** **Research Practicum Part 1 & 2 / Internship Part 1 & 2**
- **Credit Rating:** 5+ 25
- **Credit Level:** 9
- **Contact Hours:** Timetabled weekly sessions

  Research Practicum/ Internship Part 2 - E.g. 2-4 hours based on group size
- **Module Lecturer:** **Check Timetable for your group supervisor**

MSc Data Analytics (Sept 24 onwards intakes only)
MSc Open Data Practice

Research Practicum Part 1/ Internship Part 1 → Research Practicum Part 2/ Internship Part 2

Research Project - Module Learning Outcomes

On successful completion of this module, students will be able to:

- LO1 Analyse, select and implement appropriate research methods and techniques
- LO2 Research and critically analyse the state of the art of a problem domain
- LO3 Propose, architect and implement an ICT solution related to the programme area
- LO4 Evaluate the solution based on identified measures
- LO5 Investigate potential future research / commercialisation possibilities
- LO6 Present and defend the research findings through a viva, artefact/product demo and research paper style report

Practicum/Internship Part 1 & 2- Module Learning Outcomes

On successful completion of this module, students will be able to:

- LO1. Propose a research question, project objectives and methodology.
- LO2. Analyse, synthesise, and critically evaluate the state of the art.
- LO3. Propose, architect, implement and evaluate an ICT solution related to the programme area.
- LO4. Investigate potential future research and invention disclosures.
- LO5. Present and defend the research findings through a viva, artefact/product demo, and report.
- LO6. Understand the ethical issues that need to be addressed when conducting research.
- LO7. Demonstrate initiative whilst working alone or part of a team, and appropriate communication and interpersonal skills.

Research Practicum/Internship Part 1 & 2- Module Learning Outcomes

On successful completion of this module, students will be able to:

- LO1. Propose a research question, project objectives and methodology.
- LO2. Analyse, synthesise, and critically evaluate the state of the art.
- LO3. Propose, architect, implement and evaluate an ICT solution related to the programme area.
- LO4. Investigate potential future research and invention disclosures.
- LO5. Present and defend the research findings through a viva, artefact/product demo, and report.
- LO6. Identify, assess, and resolve the ethical issues that need to be addressed when conducting research.
- LO7. Demonstrate initiative whilst working alone or part of a team, and appropriate communication and interpersonal skills.

Objectives

- **To consolidate the knowledge and skills acquired in other modules by carrying out an interdisciplinary research project that combines both research and technical skills**
- **To develop an in-depth understanding of some specific topics related to their chosen discipline,**
- **To apply research methods and techniques and to develop a software/product/artefact.**
- **To investigate, design, produce, evaluate and critically analyse an innovative ICT solution related to the course area**

- **Research Project:** The research work shall build upon the work conducted as part of Research in Computing module
- **Practicum/ Research Practicum:** The research work shall build upon the work conducted as part of Practicum/Research Practicum Part 1
- **Internship:** The research work shall build upon the tasks conducted in the company. A new research proposal is required

Research Project/ Internship/ Practicum - Requirements

- **Portfolio submission consisting of**
  - Weekly activity reports required for MSc Cloud Computing, MSc Cybersecurity, MSc Fintech, MSc AI/AIBUS, MSc Data Analytics (Sept 24 intake onwards), MSc Open Data Practice
    - NOT REQUIRED for MSc Data Analytics (Jan 23/Sept 23/Jan 24 intake) doing Research Project
  - Research paper style report (up to 20 pages)
  - ICT Solution
  - Configuration manual
  - Presentation Video + Demo Video
  - Answers for the questions given by the two examiners
- **Videos**
  - Oral Presentation (10 minutes): Learners are required to record a presentation of their research
  - Demo (5-10 minutes): A demonstration of ICT solution to be illustrated

Student must submit the video recordings

Submission deadline available on MSc project Moodle page

NOTE: The College reserves the right to request a live viva session to take place in the event where examiners feel that further clarification is required regarding a student submission.

Marking Scheme – Research Project

MSc Data Analytics (Jan 23/Sept 23/Jan 24 intake) doing Research Project

PROJECT COMPONENT	WEIGHT
Project specification	10%
Literature review	10%
Artefact/ Project Development	30%
Artefact/ Product evaluation, conclusion and future work	25%
Configuration manual, document presentation, and references	15%
Viva	10%
Total	100%

Marking Scheme – Practicum/ Internship

MSc AI, MSc AI for Business, MSc Fintech, MSc Cybersecurity

Assessment Type	Assessment Description	% of total	Assessment Date
Practicum Part 1/ Internship Part 1	Project Proposal and Ethics Approval	10%	Semester 2
Practicum Part 2/ Internship Part 2	Project Specification	5%	Semester 3
	Literature Review	10%	Semester 3
	Artefact/Product Development	30%	Semester 3
	Artefact/Product Evaluation and Analysis	25%	Semester 3
	Document Presentation/Structure, Referencing, and Configuration Manual	10%	Semester 3
	Viva	10%	Semester 3

Completed

- **Weekly activity reports :** Students must provide in the supervision meetings a summary of weekly research activities performed on the project over the entire semester. These reports must be submitted on Moodle page, weekly

Marking Scheme – Research Practicum/ Internship

MSc Data Analytics (Sept 24 intake and onwards)
MSc Open Data Practice

Assessment Type	Assessment Description	% of total	Assessment Date
Research Practicum Part 1/ Internship Part 1	Project Proposal and Ethics Approval	10%	Semester 2
Research Practicum Part 2/ Internship Part 2	Project Specification	5%	Semester 3
	Literature Review	10%	Semester 3
	Artefact/Product Development	30%	Semester 3
	Artefact/Product Evaluation and Analysis	25%	Semester 3
	Document Presentation/Structure, Referencing, and Configuration Manual	10%	Semester 3
	Viva	10%	Semester 3

Completed

- **Weekly activity reports :** Students must provide in the supervision meetings a summary of weekly research activities performed on the project over the entire semester. These reports must be submitted on Moodle page, weekly.

Marking Scheme – Research Project

MSc Cloud Computing ONLY

Assessment Description	% of total
Weekly progress monitoring	12%
Project Final Submission	88%

- **Weekly progress monitoring assessment:** Students must provide a summary of weekly research activities performed on the project during the entire semester. These report must be submitted on Moodle page, weekly.

Marking Scheme – Research Project

MSc Cloud Computing ONLY

Project Final Submission Components	Weight
Project specification	5%
Literature review	8%
Artefact/ Project Development	27%
Artefact/ Product evaluation, analysis, conclusion, and future work	25%
Document presentation, ,structure and referencing	8%
User configuration manual	5%
Viva	10%
Total	88%

Note: Weekly progress reports – 12%

Ethics Consideration

- **To be considered when your project falls into one scenario or two of them or all 3 of them**

Scenario 1

Human participants are involved in your study

- Requirement specification stage
- App Interface design feedback
- Evaluation and Testing stage
- Surveys and Interviews

Declaration of Ethics Consideration Form

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Ethics Application Form

- **Primary data** – data collected by the investigator conducting the research/work (e.g. measurements, interviews, tests)

Ethics Consideration

Scenario 2

Secondary Dataset(s) is used in your study

- Data is/was collected by someone
- Data is taken from a web site(s)
- Data is provided by a company

Declaration of Ethics Consideration Form

Ethics Consideration

Scenario 2

Secondary Dataset(s) is used in your study

Scenario 2 A

PUBLIC Secondary Dataset(s) is used

- *A letter/email from the copyright holder is required to confirm potential use of data*
- *OR, Citation and link to the web site where permission is granted*

Scenario 2 B

PRIVATE Secondary Dataset(s) is used

- *Permission to use the dataset MUST BE received from the owner*
- *An approval letter/email from the owner must be attached to the form*

Ethics Consideration

Scenario 3

Target System(s) are involved in your study

- Target systems planed to be used during testing
- Real public systems are intended to be used
- Target system was not created by the student

- **Target System =** any asset, device, infrastructure component, hosting environment, network, or application, that is intended to be used by the current researcher as the target system for testing their ICT solution artefact. This can include a penetration testing target or any system under test.

Declaration of Ethics Consideration Form

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Letter/email evidence that permission/authorization has been granted to the student for using the system.

Rules of engagement evidence for penetration testing, vulnerability scans, or other ethical hacking engagement,.

REMEMBER!

It is students' responsibility to ensure that they have the correct permissions/authorizations to use any data/ target system in a study.

Projects that make use of data or target systems that does not have authorization to be used, will not be graded for that portion of the study that makes use of such data.

How to Submit my Form(s) ?

- **The following information regarding Ethics is available on Project Moodle page**
  - Declaration of Ethics Consideration Form
  - Ethical Review Application Form
  - An example of an Ethics Form submitted for a project
  - Forms submission links
- **Note**: *Provide all the required information into the Ethics Application Form (e.g. consent form, survey questions, interview questions,, details on how the collected data will be analysed, etc.)*

- **Submission link**: Available on MSc project Moodle page

If ethical approval was received in Research in Computing/ Practicum Part 1/ Research Practicum Part 1 and if there are no changes regarding ethical conditions, there is no need for resubmission of the form.